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Project #25: Putting a CAP on Discharge Antimicrobial Therapy: Evaluation of a Transitions of Care Process Through an Electronic Scoring System for Patients with Community-Acquired Pneumonia (CAP) and Chronic Obstructive Pulmonary Disease (COPD)

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AIM: Combat Discharge Antibiotic Overuse

Problem

- Less than 1 in 5 hospitals with monitor antibiotic use at hospital discharge, leading to antibiotic overuse.
- Antimicrobial Stewardship Program Transition of Care (ASP TOC) is a standard of care pharmacist-driven intervention: pharmacists pend discharge antimicrobial orders and write progress notes for patients with uncomplicated infections being discharged on antibiotics.
- Identification of ASP TOC opportunities through an electronic scoring system the electronic health record (EHR) may allow for earlier and more frequent identification of pharmacist intervention.

Aim

- To evaluate a systematic process using an electronic variable (ASP TOC LRTI variable) in the electronic scoring system to identify patients eligible for ASP TOC interventions in patients diagnosed with lower respiratory tract infections (LRTI)

Pre-Group: ASP TOC Patient Identification

- Manual patient roster review to determine eligibility
- Pharmacist workload of ~40-50 patients

Post-Group: ASP TOC Patient Identification

- ESS implementation included ASP TOC LRTI variable
- ESS identification and triage for eligible patients
- Pharmacist workload of ~60 patients

PLAN: Current State, Barriers, Solution

ASP TOC Impact: Pharmacist-driven ASP TOC intervention increased optimal antibiotic prescribing by 46% (Mercuro et al., JAMA 2022)

Barriers

- Inconsistent methods in identification of patients eligible for ASP TOC intervention
- Inconsistent staffing models within sites, off-hours staffing

Potential Solution

- Implementation of ASP TOC LRTI variable on the electronic scoring system may help identify patients that can be discharged with antibiotics

Figure 1. ASP TOC LRTI variable in electronic scoring system on EHR

Figure 2. ASP TOC LRTI variable criteria. Patients must be prescribed one of the listed antibiotics for CAP or COPD for greater than 48h.

ASP TOC LRTI Variable

- Antibiotics: ceftriaxone, doxycycline, azithromycin, ampicillin-sulbactam, moxifloxacin
- Antibiotic therapy > 48 hours for indication: LRTI

DO: ASP TOC LRTI Intervention & Assessment

Figure 3. Timeline of Electronic Scoring System and ASP TOC LRTI Implementation

1. ***October 2021*** — Electronic Scoring System (with variables) launches on EHR
2. Standard of care for streamlining pharmacy services
3. ***September 2022*** — Intervention: **ASP TOC LRTI variable** launches on EHR
4. ***Sept 2022 – March 2023*** — ASP TOC LRTI variable in use. Data collection in process.

Study Implementation

- IRB-approved, retrospective, quasi-experimental study on patients discharged from 11/1/21-3/1/22 and from 11/1/22-3/1/23 diagnosed with CAP or COPD

Education and Screening

- Educated physicians on ASP TOC process, roles of RPh and MD
- Education sessions with pharmacy residents and students on ASP TOC process and how to leverage ASP TOC LRTI variable for identification
- Distribution of ASP TOC pocket cards, which included empiric therapy and duration for selected uncomplicated infections

CHECK: Compare Pre-Intervention Group to Post-Intervention Group

Comparison of pre-intervention vs post-intervention group:

- Primary Endpoint: Proportion of optimized discharge antimicrobial regimens
- Secondary Endpoint:
  - Proportion of pharmacist-led ASP TOC interventions (indicated by pharmacist-written ASP TOC progress note and pended discharge antibiotic order)
  - Outcomes associated with optimized discharge antimicrobial regimens

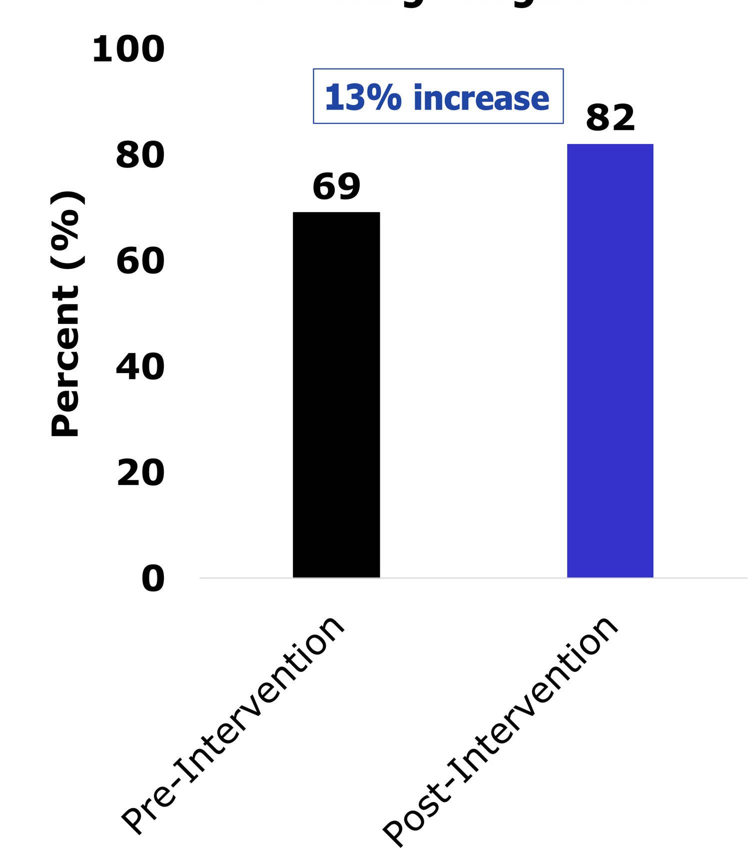
Analysis

- Sample size: 200 patients (100 per intervention group)
- Descriptive statistics, Mann-Whitney U test, T-test, multivariate logistic regression

MEASURES: Optimized Discharge Regimens Improved with ASP TOC LRTI Intervention

Table 1. Baseline Characteristics

Baseline Characteristics	Pre-Intervention (N = 100)	Post-Intervention (N = 100)	P-value
Male gender (n, %)	46 (46)	49 (49)	0.671
Age (years, IQR)	66.9 (56.5-72.7)	67.0 (58.4-75.3)	0.394
Comorbidities (n, %)			
Asthma	22 (22)	32 (32)	0.111
COPD	52 (52)	61 (61)	0.199
Smoking History	62 (62)	75 (75)	0.048
Non-equivalent Dependent Variable: Deep Vein Thrombosis (DVT) Prophylaxis (n, %)			
No DVT Prophylaxis	26 (26)	28 (28)	0.355
DVT Prophylaxis present	74 (74)	72 (72)	
Risk Factors for Multi-drug Resistant Organisms (MDRO) (n, %)			
Hospitalization ≥ 2 days and IV antibiotics within the last 90 d	13 (13)	10 (10)	0.506
MDRO within 365 days	3 (3)	2 (2)	0.651



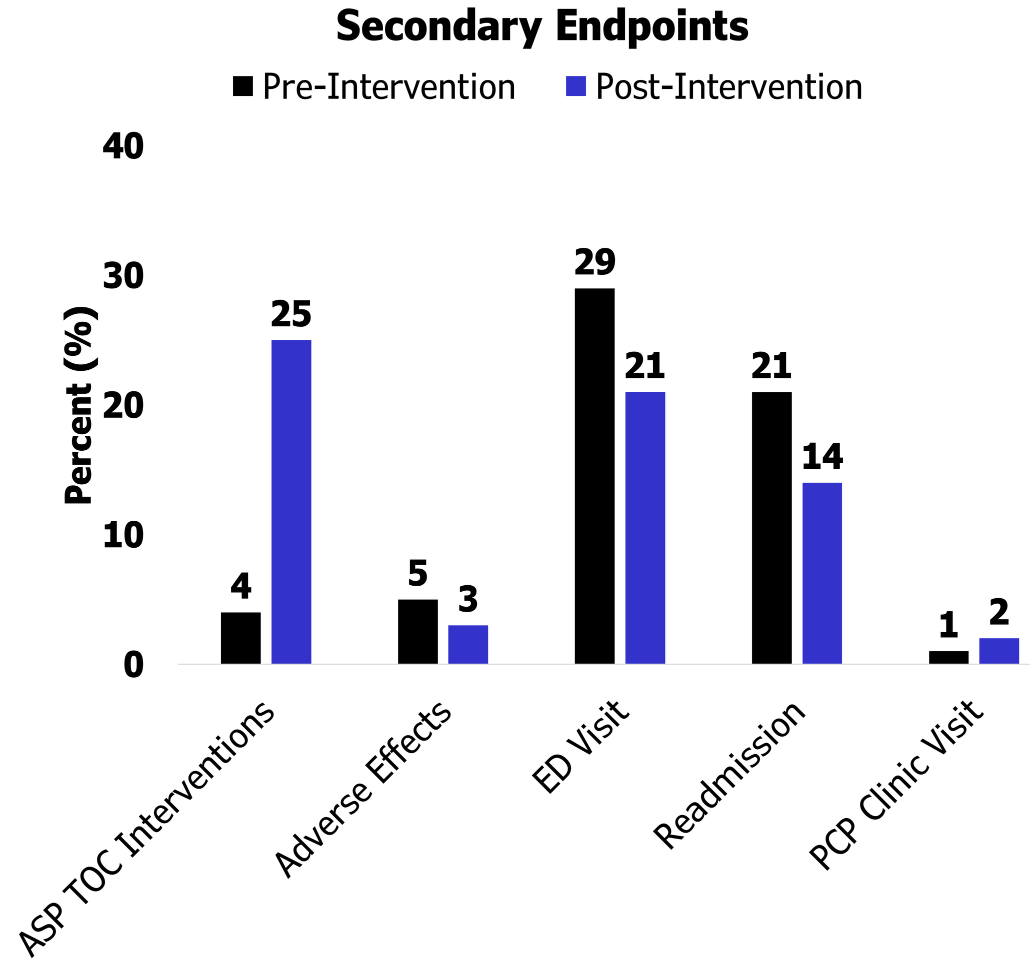


Table 2. Factors Associated with Optimized Antimicrobial Discharge Regimens

Factor	Adjusted Odds Ratio	P-Value	95% Confidence Interval
ASP TOC Intervention	6.57	0.012	1.51-28.63
Female Gender	1.61	0.172	0.81-3.17
COPD	3.89	< 0.001	1.85-8.20

ACT: Sustain and Spread

Sustain

- Use of the electronic scoring system and ASP TOC LRTI variable can aid in identifying patients that require antimicrobial therapy at discharge and who may be eligible for ASP TOC intervention.

Spread

- The use of an ASP TOC scoring variable for other uncomplicated infections (e.g. urinary tract infections, intra-abdominal infections, skin/soft tissue infections) can be also considered in the future.
- Findings of this study have been presented at local and national conferences

KEYS TO SUCCESS / LESSONS LEARNED

Lessons Learned: The ASP TOC LRTI variable has aided pharmacists in identification of patients with LRTI, allowing for further collaboration with the physician teams, and improving discharge antibiotic duration and selection.

Keys to Success: Multidisciplinary collaboration within pharmacy departments (infectious diseases, transition of care, inpatient, outpatient) and medical teams was essential for the success of this intervention.

- The use of artificial intelligence, such as the use of the electronic scoring system, can be leveraged by clinical pharmacists to maintain high-quality patient care.